Hauya elegans (Guayabón) at Huaravito Reserve LTEMS: Population Structure and Potential Role as an Old-Growth Indicator Species

Technical Species Report

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Executive Summary

Seventy-two observations of Hauya elegans (Onagraceae), locally known as Guayabón or Supara, have been documented via iNaturalist within the Toyopán Sacred Forest at Huaravito Reserve LTEMS. This species, historically dominant in the premontane forests of Costa Rica's Central Valley prior to widespread deforestation, is now regionally rare, with extant populations largely restricted to steep ravines and protected remnants. The high recurrence of large-diameter H. elegans individuals at Huaravito, combined with field evidence of deep organic soil layers, absence of compaction, and continuous forest cover since at least 2000, suggests that the site may harbor a relict old-growth population of this ecologically and structurally significant tree. This report synthesizes available species information, describes the Huaravito population, and discusses the conservation implications of this finding.

1. Species Background: Hauya elegans

1.1 Taxonomic and Morphological Description

Hauya elegans D.C. (family Onagraceae) is a medium to large canopy tree native to regions from southern Mexico through Central America to Panama. In Costa Rica, it occurs in premontane humid zones, particularly in the Central Valley, where it is known by several common names: Guayabón, Supara (also Guayabón Supara), Palo Borrego, and Palo de Ajo.

Note on the name "Supara":

The etymology of "Supara" remains uncertain and appears to be a uniquely Costa Rican vernacular term, used primarily in the Central Valley region. Despite phonetic characteristics that might suggest indigenous origin, searches in documented Huetar and Bribri lexicons have not identified "Supara" as a recorded word in these languages. It is more likely a local colloquial name that developed within rural Spanish-speaking communities of the Central Valley, possibly influenced by regional phonetic patterns rather than direct indigenous etymology. The name is particularly associated with areas where H. elegans was historically abundant in old-growth premontane forests that have since been largely converted to urban and agricultural uses. Its preservation in local oral tradition underscores the biocultural significance of the species in the collective memory of Central Valley communities.

The species is characterized by:

- Large buttressed roots (“trunks with legs”) providing structural stability on slopes
- Smooth gray bark with lenticels
- Simple alternate leaves, elliptic to obovate, 8–15 cm long
- Small greenish-white flowers in terminal panicles
- Fleshy capsular fruits 1–2 cm diameter, consumed by birds and mammals

Mature individuals can reach 25–30 meters in height and >80 cm diameter at breast height (DBH). The species is shade-tolerant in its juvenile stages but becomes a canopy emergent in mature forests. It is closely associated with premontane and lower montane life zones (800–1,800 m.a.s.l.) in Costa Rica.

1.2 Historical Distribution and Current Conservation Status

According to regional natural history sources (El Mundo Forestal, 2024), H. elegans was once a dominant canopy species in the primary premontane forests of Costa Rica’s Central Valley, particularly in the Heredia-Alajuela-San José uplands. Extensive deforestation during the 19th and 20th centuries for coffee plantations, cattle pasture, and urban expansion resulted in near-total loss of these forests. Today, H. elegans is considered uncommon to rare across its former range in the Central Valley.

Remaining populations are largely confined to:

- Steep ravines and riparian corridors (e.g., Río Virilla barrancas) where topography prevented agricultural conversion
- Protected areas and forest reserves (e.g., Braulio Carrillo National Park, private reserves)
- Isolated old-growth patches embedded in agricultural matrices

While not formally listed as threatened under IUCN criteria (likely due to its broader Central American distribution), H. elegans is functionally rare in many parts of its historical Costa Rican range. The loss of large-diameter reproductive individuals and fragmentation of populations may limit genetic diversity and long-term viability in some regions.

2. Hauya elegans Population at Huaravito Reserve LTEMS

2.1 Documentation and Sampling Effort

Between 2021 and 2026, 72 individual H. elegans trees were observed and photographically documented within the Huaravito Reserve (5,800 m² total area; ~4,000 m² forested core). All observations were uploaded to the iNaturalist platform (Project: Reserva Huaravito / Sitio Natural Sagrado) with GPS coordinates and taxonomic verification by the global iNaturalist community.

Important methodological note:

The 72 observations represent detected individuals during opportunistic field surveys, not a systematic census with complete spatial coverage. Therefore, these data likely reflect: (a) preferential sampling of large, conspicuous individuals; (b) spatial clustering in easily accessible areas; and (c) underestimation of total population size (juveniles and small trees may be underrepresented). As such, the count should be interpreted as a minimum confirmed abundance rather than a definitive population census.

2.2 Population Structure

Field observations indicate that the documented H. elegans population is characterized by:

- Predominance of large-diameter individuals: Many trees exhibit extensive buttress roots and DBH visually estimated at >50 cm (formal dendrometric measurements pending).
- High epiphytic colonization: Trunk surfaces show >70% coverage by bryophytes, orchids (including Scaphyglottis spp.), and ferns, indicating stable, humid microclimate conditions and structural longevity.
- Spatial concentration in western sector: The majority of observations cluster in the area designated as “The Shrine,” corresponding to the mature forest core with the deepest organic soil layers and highest canopy closure.
- Presence across canopy strata: While most documented individuals are canopy or sub-canopy trees, occasional saplings and pole-stage individuals suggest ongoing (albeit limited) recruitment.

2.3 Ecosystem Role

At Huaravito, H. elegans is structurally prominent and functionally significant:

- Canopy architecture: Provides upper canopy structure that intercepts rainfall, reduces wind velocity, and creates shaded understory conditions favorable for shade-tolerant species.
- Epiphyte substrate: The rough bark and long-lived trunks support diverse epiphytic communities, contributing to site-level biodiversity.
- Thermal regulation: The dense canopy and high epiphytic biomass likely contribute to the exceptional biogenic cooling effect (10.2°C gradient) documented at the site, as mature H. elegans trees create the structural complexity necessary for evapotranspirative cooling and wind attenuation.
- Faunal resources: Fruits provide food for birds and mammals; large trunks with cavities may serve as nesting or roosting sites.

3. Evidence for Old-Growth Status: The Toyopán Patch

3.1 Convergent Lines of Evidence

While definitive proof of old-growth status requires dendrochronological analysis (tree ring dating) or historical land-use documentation, multiple independent observations are consistent with the hypothesis that the Toyopán Sacred Forest represents a minimally disturbed old-growth remnant:

A. Continuous Forest Cover (Satellite Evidence)

Global Forest Watch data indicate canopy cover >30% at the site as early as 2000 (the earliest available high-resolution imagery). While 30% cover does not necessarily imply mature forest (it could represent young secondary growth), it does establish that the site was not open pasture or cropland in 2000. Oral testimony from local residents (50+ years) corroborates continuous forest presence, though the condition (primary vs. secondary) prior to 1984 remains undocumented.

B. Soil Profile Characteristics

Field excavation of soil pits within the H. elegans-dominated zone revealed:

- Deep organic layer (~20 cm): Composed of decomposing leaf litter, fine roots, and fungal hyphae, characteristic of long-term forest conditions.
- Gradual horizonation: No abrupt transitions between organic and mineral layers, suggesting continuous organic matter accumulation over decades to centuries.
- Absence of compaction: No plow pan, no hardpan, no compressed layers indicative of past agricultural use.
- High bioturbation: Extensive root networks, earthworm activity, and macrofauna presence indicate active, undisturbed soil biological processes.
- Charcoal presence: Isolated charcoal fragments at depth do not contradict old-growth status; fire may have been part of the long-term ecological history of the site (natural or indigenous low-intensity burning).

C. Tree Size and Structural Complexity

The presence of multiple large-diameter H. elegans individuals (>50–80 cm DBH estimated) with well-developed buttress roots is inconsistent with young secondary forest. Such structural features require decades to develop. Combined with high canopy closure, multi-layered vertical structure, and dense epiphytic loading, the forest exhibits the physiognomy of mature or old-growth premontane forest.

3.2 Working Interpretation

Based on the convergent evidence above, the following interpretation is proposed:

> *The Toyopán Sacred Forest likely represents an undisturbed or minimally disturbed old-growth premontane forest patch. While absolute certainty requires dendrochronological confirmation, the cumulative evidence — continuous forest cover since at least 2000, deep organic soils with no agricultural disturbance signature, large-diameter trees with mature structural attributes, and high ecosystem complexity — is consistent with old-growth status. The site may represent one of the last intact remnants of the premontane forest matrix that once dominated the Alajuela-San Ramón uplands.*

This interpretation positions Huaravito not as a "recovering secondary forest" but as a mature forest nucleus functioning as a thermal and ecological refugium within a fragmented agricultural landscape.

4. Conservation Significance

4.1 Regional Rarity of Hauya elegans

If the Huaravito population represents 72+ mature individuals within <0.6 hectares, the site may harbor one of the densest concentrations of H. elegans currently documented in the Central Valley region. For comparison, remnant populations in Río Virilla ravines are described as "isolated ancient individuals," suggesting low densities and fragmented distributions. The demographic structure at Huaravito — multiple size classes, spatial clustering, and ongoing (if limited) recruitment — may represent a more demographically viable population than scattered individuals in degraded matrix landscapes.

4.2 Old-Growth Refugium Hypothesis

If the Toyopán patch is confirmed as old-growth, its conservation value extends beyond H. elegans to include:

- Genetic repository: Preservation of genotypes adapted to historical climatic and biotic conditions, potentially valuable for restoration efforts in a changing climate.
- Ecological baseline: A reference condition for understanding premontane forest structure, function, and species composition prior to widespread anthropogenic modification.
- Climate refugium: The documented 10.2°C cooling effect suggests the site provides thermal microrefugia for heat-sensitive species under regional warming scenarios.
- Biocultural heritage: The site's designation as a Sacred Natural Site (Toyopán) reflects its spiritual and cultural significance, adding a biocultural dimension to its conservation importance.

4.3 Threats and Vulnerabilities

Despite its apparent old-growth status and current protection under custodial management, the Huaravito H. elegans population faces potential threats:

- Small population size: Even with 72+ individuals, the total population may be vulnerable to stochastic events (windthrow, disease, drought).
- Edge effects: The site's irregular geometry and proximity to agricultural lands (clay extraction site, pastures, roads) expose the forest to edge-related stressors (wind, altered hydrology, invasive species).
- Limited recruitment: While some regeneration is observed, the dense canopy and thick organic layer may limit seedling establishment. Long-term demographic viability depends on successful recruitment to replace senescent trees.
- Climate change: Shifts in temperature and precipitation regimes may alter the microclimate stability that currently supports the population.

5. Recommendations

5.1 Population Monitoring

- Conduct a systematic census with complete spatial coverage to determine total population size and demographic structure (seedlings, saplings, poles, adults).
- Establish permanent monitoring plots to track recruitment, growth, and mortality over time.
- Measure DBH of all individuals ≥10 cm to quantify size distribution and estimate biomass.
- Map spatial distribution using GPS to identify clustering patterns and potential microhabitat associations.

5.2 Dendrochronological and Genetic Studies

- Extract increment cores from large individuals to determine age structure and confirm old-growth status.
- Conduct genetic analysis (microsatellites or SNPs) to assess genetic diversity and compare with other H. elegans populations in Costa Rica.

5.3 Habitat Protection and Management

- Maintain current custodial management regime (no logging, no understory clearing, minimal human disturbance).
- Monitor and control invasive species (Pennisetum purpureum, exotic vines) at forest edges to prevent encroachment.
- Consider buffer zone expansion or corridor creation to reduce edge effects and enhance connectivity with other forest patches.

5.4 Collaboration and Publication

- Partner with academic institutions (UCR, UNA, OTS) to conduct formal demographic and ecological studies.
- Publish population data and old-growth hypothesis in peer-reviewed journals (e.g., Brenesia, Revista de Biología Tropical).
- Contribute data to national and regional biodiversity databases (INBio, GBIF) to inform conservation planning.

6. Conclusion

The 72 documented observations of Hauya elegans at Huaravito Reserve LTEMS represent a significant finding in the context of regional forest conservation. This species, once dominant in the premontane forests of the Central Valley but now rare due to extensive deforestation, persists at Huaravito in what appears to be a structurally intact, demographically viable population. The convergent evidence — large-diameter individuals, deep undisturbed soils, continuous forest cover, and high ecosystem complexity — suggests that the Toyopán Sacred Forest may be a relict old-growth remnant rather than recovering secondary forest.

If this interpretation is correct, Huaravito represents not only an important site for H. elegans conservation but also a rare example of a mature premontane forest refugium embedded in an agricultural landscape. The site's exceptional thermal stability (10.2°C cooling effect) and high biodiversity (116 species documented) further underscore its ecological and scientific value. Systematic demographic monitoring, dendrochronological analysis, and collaborative research are recommended to confirm old-growth status and inform long-term conservation strategies.

References

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